$TmFeO_3$ Hamiltonian and Tm Magnetization Readout

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This note summarizes the Hamiltonian implemented in `src/core/unitcell_builders.cpp` for `build_tmfeo3()`, using the notation and physical definitions of the companion theory document `tmfeo3_notes.tex`. Section 5 records how the effective Tm SU(3) variables are transformed when magnetization is saved, and Section 6 details the time-dependent drive.

Contents

1 Conventions

1.1 Sites and sublattice frames

The mixed $TmFeO_3$ unit cell contains four Fe sites and four Tm sites. Fe spins are three-component SU(2) vectors $\mathbf{S}_i = (S_i^x, S_i^y, S_i^z)$ with spin length S; Tm sites are stored as eight-component real Gell-Mann expectation-value vectors $\boldsymbol{\lambda}_j = (\lambda_1, \ldots, \lambda_8)$ in the local crystal-field (CEF) basis.

The four Pbnm sublattices are related by the Klein four-group of proper π-rotations $\{R_\mu\}_{\mu=0}^{3} = \{E, C_{2x}, C_{2y}, C_{2z}\}$, represented as diagonal matrices

$$R_0 = \mathrm{diag}(+,+,+), \quad R_1 = \mathrm{diag}(+,-,-), \quad R_2 = \mathrm{diag}(-,+,-), \quad R_3 = \mathrm{diag}(-,-,+). \tag{1}$$

In the code, the same matrices are written as $D_\mu = \mathrm{diag}(\eta_{\mu,x}, \eta_{\mu,y}, \eta_{\mu,z})$. A global spin is recovered from the local-frame spin by $S_\mu^\alpha\big|_{\mathrm{glob}} = (R_\mu)_{\alpha\alpha}\, \tilde{S}_\mu^\alpha\big|_{\mathrm{loc}}$.

1.2 Bertaut modes

The four sublattice combinations

$$\mathbf{X}_{\mathbf{n}} = \tfrac{1}{4} \sum_{\mu=0}^{3} \sigma_\mu^X\, \mathbf{S}_{\mathbf{n},\mu}, \quad X \in \{F, G, C, A\}, \tag{2}$$

with sign vectors

$$\sigma^F = (+,+,+,+),\ \sigma^G = (+,-,-,+),\ \sigma^C = (+,+,-,-),\ \sigma^A = (+,-,+,-), \tag{3}$$

span the Bertaut basis. The inverse relation is $\mathbf{S}_{\mathbf{n},\mu} = \sum_X \sigma_\mu^X\, \mathbf{X}_{\mathbf{n}}$. The Γ_2 ground state has $\mathbf{G} = G_z\hat{z}$, $\mathbf{F} = F_x\hat{x}$, $\mathbf{C} = C_y\hat{y}$, $\mathbf{A} \approx 0$.

1.3 Gell-Mann classification

The eight Gell-Mann generators are classified by mirror parity (m_z, site symmetry C_s) and time-reversal parity T:

$$\boxed{\begin{array}{rl} A_1^+ \text{ (mirror-even, } T\text{-even)}: & \lambda^1, \lambda^3, \lambda^8, \\ A_1^- \text{ (mirror-even, } T\text{-odd)}: & \lambda^2, \\ A_2^+ \text{ (mirror-odd, } T\text{-even)}: & \lambda^4, \lambda^6, \\ A_2^- \text{ (mirror-odd, } T\text{-odd)}: & \lambda^5, \lambda^7. \end{array}} \tag{4}$$

The T-odd generators $\lambda^2, \lambda^5, \lambda^7$ carry the projected magnetic dipole $\mathbf{J}^{\text{eff}}$; the T-even off-diagonal generators $\lambda^1, \lambda^4, \lambda^6$ carry the projected electric dipole $\mathbf{d}^{\text{eff}}$.

1.4 Projected dipole operators

In the T-symmetric gauge the three lowest Tm CEF singlets $\{|E_1(A_1)\rangle, |E_2(A_1)\rangle, |E_3(A_2)\rangle\}$ have numerical matrix elements (see `tmfeo3_notes.tex` §III.C)

$$\mathbf{J}^{\text{eff}} = g_J\mu_B \begin{pmatrix} J_z^{\text{eff}} \\ J_x^{\text{eff}} \\ J_y^{\text{eff}} \end{pmatrix} = g_J\mu_B \begin{pmatrix} +5.264\,\lambda^2 \\ +2.392\,\lambda^5 + 0.913\,\lambda^7 \\ -2.787\,\lambda^5 + 0.466\,\lambda^7 \end{pmatrix}, \tag{5}$$

which defines the 3×8 active-channel matrix μ_{act} used in the code:

$$\mu_{\text{act}} = \begin{pmatrix} 0 & \mu_{2x} & 0 & 0 & \mu_{5x} & 0 & \mu_{7x} & 0 \\ 0 & \mu_{2y} & 0 & 0 & \mu_{5y} & 0 & \mu_{7y} & 0 \\ 0 & \mu_{2z} & 0 & 0 & \mu_{5z} & 0 & \mu_{7z} & 0 \end{pmatrix}, \tag{6}$$

so that $(J^\alpha)^{\text{eff}} = \sum_{a\in\{2,5,7\}} \mu_{\alpha a}\lambda^a$. The projected electric dipole $\mathbf{d}^{\text{eff}}$ populates λ^1 (via $d_{x,y}$) and $\lambda^{4,6}$ (via d_z).

2 Fe Sector

2.1 Four-sublattice Hamiltonian

The Fe sector is the standard four-sublattice orthoferrite Hamiltonian with Heisenberg exchange, Dzyaloshinskii–Moriya (DM) interaction, and biaxial single-ion anisotropy:

$$
\begin{aligned}
H_{\mathrm{Fe}} = & \sum_{\langle ij\rangle_\parallel} J_1\,\mathbf{S}_i\cdot\mathbf{S}_j + \sum_{\langle ij\rangle_c} J_{1c}\,\mathbf{S}_i\cdot\mathbf{S}_j + \sum_{\langle\langle ij\rangle\rangle} J_2\,\mathbf{S}_i\cdot\mathbf{S}_j \\
& + \sum_{\langle ij\rangle_\parallel} \mathbf{D}_{ij}\cdot(\mathbf{S}_i\times\mathbf{S}_j) \\
& - \sum_i \big[K_a(S_i^x)^2 + K_b(S_i^y)^2 + K_c(S_i^z)^2\big].
\end{aligned}
\tag{7}
$$

Here J_1 (`J1ab`), J_{1c} (`J1c`), J_2 (`J2ab`/`J2c`) are the in-plane NN, c-axis NN, and NNN exchange constants; $K_{a,b,c}$ (`Ka`, `Kb`, `Kc`) are the biaxial single-ion anisotropy constants.

2.2 DM bond pattern

The DM vectors on in-plane NN bonds are fixed by *Pbnm* symmetry. With bonds at height $z=\frac{1}{2}$ and $z=0$ distinguished by the S_1 screw axis:

$$
\mathbf{D}_{ij}^{(z=\frac{1}{2})} = (0,+D_1,+D_2), \qquad \mathbf{D}_{ij}^{(z=0)} = (0,-D_1,+D_2). \tag{8}
$$

The sign reversal of D_1 between the two planes is required by the S_1 screw that maps one z-plane to the other; D_2 is symmetric.

Code implementation

The code encodes each bond as a lab-frame exchange matrix via

$$
J(J_{\mathrm{iso}}, D_y, D_z) = \begin{pmatrix} J_{\mathrm{iso}} & D_z & -D_y \\ -D_z & J_{\mathrm{iso}} & 0 \\ D_y & 0 & J_{\mathrm{iso}} \end{pmatrix}. \tag{9}
$$

With the parameter names in `apply_tmfeo3_fe_sector()`, the NN exchange matrices assigned to each bond set are

$$
\begin{aligned}
J_a^{(10)} = J_b^{(10)} &= J(J_{1ab}, +D_1, +D_2), \\
J_a^{(23)} = J_b^{(23)} &= J(J_{1ab}, -D_1, +D_2), \\
J_c,\ J_{2a},\ J_{2b},\ J_{2c} &= J_{\mathrm{iso}}\,\mathbb{1}_{3\times 3}.
\end{aligned}
\tag{10}
$$

The antisymmetric part of J_{ij} is identified with $\mathbf{D}_{ij}\cdot(\mathbf{S}_i\times\mathbf{S}_j)$, reproducing Eq. (8).

2.3 Fe Zeeman (static field)

A static external field $\mathbf{h} = h\,\hat{n}$ adds

$$H_{\mathrm{Fe,Z}} = -g_e\mu_B \sum_i \mathbf{h}\cdot\mathbf{S}_i, \tag{11}$$

where in the code the $g_e\mu_B$ prefactor is absorbed into `field_strength` $=$ $g_e\mu_B\,h$ and `field_direction` $= \hat{n}$.

3 Tm Sector

3.1 CEF Hamiltonian: three-level truncation

The microscopic Tm^{3+} CEF Hamiltonian is a Stevens-operator sum over even-$|m|$ terms allowed by the C_s site symmetry, truncated to the lowest three singlets $\{|E_1(A_1)\rangle, |E_2(A_1)\rangle, |E_3(A_2)\rangle\}$ with energies $\epsilon_1 = 0$, $\epsilon_2 \equiv e_1$, $\epsilon_3 \equiv e_2$ (code parameters `e1`, `e2`). In the truncated qutrit basis the diagonal part is

$$H^{(0)}_{\mathrm{CEF}} = \bar{\epsilon}\,\mathbb{1} + h^{(0)}_3\lambda^3 + h^{(0)}_8\lambda^8, \tag{12}$$

with

$$h^{(0)}_3 = \frac{e_1}{2}, \qquad h^{(0)}_8 = \frac{e_1 - 2e_2}{2\sqrt{3}}, \tag{13}$$

and $\bar{\epsilon} = (e_1 + e_2)/3$ irrelevant. Numerically, $h^{(0)}_3 = -0.96\,\mathrm{meV}$, $h^{(0)}_8 = -3.97\,\mathrm{meV}$. The code stores the on-site SU(3) field vector

$$\mathbf{B}_{\mathrm{CEF}} = (0, 0, \alpha, 0, 0, 0, 0, \beta), \tag{14}$$

with

$$\alpha = -h^{(0)}_3\,s_\alpha, \qquad \beta = -h^{(0)}_8\,s_\beta, \tag{15}$$

and scale factors `tm_alpha_scale` $= s_\alpha$, `tm_beta_scale` $= s_\beta$ (default 1). The on-site Tm CEF contribution per site is $H^{(0)}_{\mathrm{CEF}} = -\mathbf{B}_{\mathrm{CEF}}\cdot\boldsymbol{\lambda}_j$.

3.2 Tm Zeeman coupling (static field)

The static field couples to the projected magnetic dipole:

$$H_{\mathrm{Tm,Z}} = -\sum_j \mathbf{J}^{\mathrm{eff}}_j \cdot \mathbf{h} = -g_{\mathrm{ratio}} \sum_j \sum_{a\in\{2,5,7\}} B_a\,\lambda^{(j)}_a, \tag{16}$$

where $B_a = \sum_\alpha \mu_{\alpha a}\,h_\alpha$ projects the field onto the active T-odd channels, and `g_ratio_tm` $= g_{\mathrm{ratio}} = g_J/g_e$ is the Tm-to-Fe g-factor ratio.

3.3 Tm–Tm diagonal bilinears

Nearest-neighbor Tm–Tm interactions are parameterized as a diagonal coupling in the Gell-Mann basis:

$$H_{\mathrm{TmTm}} = \frac{1}{2} \sum_{\langle j,j' \rangle} \sum_{a=1}^{8} J_a^{\mathrm{Tm}} \, \lambda_a^{(j)} \lambda_a^{(j')}, \tag{17}$$

implemented via the 8×8 diagonal matrix $J_{\mathrm{Tm}} = \mathrm{diag}(J_{\mathrm{tm},1}, \ldots, J_{\mathrm{tm},8})$ on the bond geometry listed in `apply_tmfeo3_tm_sector()`.

4 Fe–Tm Couplings

4.1 Bilinear Fe–Tm coupling (χ term)

Because the Hamiltonian must be T-even and the Fe spin $\mathbf{S}$ is T-odd, only the T-odd Gell-Mann generators $\lambda^{2,5,7}$ can appear linearly (Eq. (4)). The most general *Pbnm*-symmetric bilinear Fe–Tm coupling is

$$H_\chi = \sum_{(i,j)} \sum_{\alpha=x,y,z} \sum_{a \in \{2,5,7\}} \chi_{ij}^{\alpha a} \, S_i^\alpha \, \lambda_j^a, \tag{18}$$

with bond-resolved coupling tensor $\chi_{ij}^{\alpha a}$. The active channels in matrix form are

$$\chi = \begin{pmatrix} 0 & \chi_{2x} & 0 & 0 & \chi_{5x} & 0 & \chi_{7x} & 0 \\ 0 & \chi_{2y} & 0 & 0 & \chi_{5y} & 0 & \chi_{7y} & 0 \\ 0 & \chi_{2z} & 0 & 0 & \chi_{5z} & 0 & \chi_{7z} & 0 \end{pmatrix}. \tag{19}$$

At $\mathbf{q} = 0$ only $\bar{\chi}^{2z} \neq 0$ survives the inversion-pair cancellation; this is the dominant channel coupling λ^2 (A_1^-) to G_z (the Γ_2 Néel order parameter). The off-diagonal mirror-odd channels $\chi^{5x,7x,5y,7y}$ cancel at $\mathbf{q} = 0$ due to the Pbnm bond-pair structure.

The code stores 16 explicit inversion-related bond pairs; on the inversion-odd partner bond the λ^5 and λ^7 channels flip sign while λ^2 does not.

4.2 Mixed trilinear W term

The T-even generators $\lambda^{1,3,4,6,8}$ do not appear in the linear coupling H_χ, but they enter at second order via virtual Tm transitions. Expanding to quadratic order in the Bertaut amplitudes, the resulting $\mathbf{q} = 0$ effective

Hamiltonian is

$$H_W = -\sum_{\mathbf{n},j}[u_1\lambda^1_{\mathbf{n},j} + u_3\lambda^3_{\mathbf{n},j} + u_8\lambda^8_{\mathbf{n},j}]\mathcal{I}^{\mathrm{Fe}}_{A_1^+}(\mathbf{n}) - \sum_{\mathbf{n},j}[v_4\lambda^4_{\mathbf{n},j} + v_6\lambda^6_{\mathbf{n},j}]\mathcal{I}^{\mathrm{Fe}}_{A_2^+}(\mathbf{n}), \tag{20}$$

where the A_1^+ (mirror-even) Fe bilinears are

$$\mathcal{I}^{\mathrm{Fe}}_{A_1^+} = \alpha_1 G_z^2 + \alpha_2 F_x^2 + \alpha_3 C_y^2 + \cdots \tag{21}$$

and the A_2^+ (mirror-odd) ones are

$$\mathcal{I}^{\mathrm{Fe}}_{A_2^+} = \beta_1 F_x G_z + \beta_2 G_z C_y + \cdots \tag{22}$$

with α_i, β_i phenomenological $\mathbf{q} = 0$ coefficients. In the code, `build_tmfeo3()` populates $n \in \{1, 3, 4, 6, 8\}$ channels of the on-site trilinear tensor $W^{\alpha\beta}_{n,ij}$. The A_2^+ channels (λ^4, λ^6) change sign on inversion-odd bond partners; the A_1^+ channels ($\lambda^{1,3,8}$) are even. The A_2^+ on-site contribution vanishes at $\mathbf{q} = 0$ in the inversion-pair cancellation, leaving the A_1^+ population-modulation channel as the dominant back-action mechanism.

4.3 Total Hamiltonian

The full working Hamiltonian is

$$\boxed{H_{\mathrm{total}}(t) = H_{\mathrm{Fe}} + H^{(0)}_{\mathrm{CEF}} + H_{\mathrm{Tm,Z}} + H_{\mathrm{TmTm}} + H_\chi + H_W + H_{\mathrm{drive}}(t),} \tag{23}$$

corresponding to the microscopic Hamiltonian $H_{\mathrm{micro}}(t)$ of `tmfeo3_notes.tex` Eq. (Hmicro) augmented by the W back-action term, Eq. (Htotal).

5 How the Tm Magnetization Is Saved

5.1 Stored local SU(3) variables

Each Tm site is evolved and stored internally as an eight-component local vector $\boldsymbol{\lambda}^{\mathrm{loc}} = (\lambda_1, \ldots, \lambda_8)$ in the canonical local CEF basis.

5.2 Sublattice-frame map for readout

The builder constructs an 8×8 sublattice frame F_μ from the projected dipole matrix:

$$F_\mu = \mu_{\mathrm{act}}^{-1}\, D_\mu\, \mu_{\mathrm{act}} \quad \text{on the active triplet } \{\lambda_2, \lambda_5, \lambda_7\}. \tag{24}$$

The same 3×3 action is extended to the coherence-partner triplet $\{\lambda_1, \lambda_4, \lambda_6\}$, while λ_3 and λ_8 are left invariant. In block form,

$$\boldsymbol{\lambda}^{\mathrm{glob}} = F_\mu\, \boldsymbol{\lambda}^{\mathrm{loc}}. \tag{25}$$

The frame is installed via `Tm_atoms.set_sublattice_frame(...)` and copied into `sublattice_frames_SU3` when the `MixedLattice` is built.

5.3 What the saved SU(3) magnetization means

When the code computes global SU(3) magnetization it applies the frame site by site:

$$M_{\mathrm{SU3},\mu}^{\mathrm{glob}} = \frac{1}{N_{\mathrm{Tm}}} \sum_{j\in\mu} F_\mu\, \boldsymbol{\lambda}_j^{\mathrm{loc}}. \tag{26}$$

- `M_local_SU3`: average local qutrit Bloch vector in the CEF basis.
- `M_global_SU3`: average SU(3) vector after F_μ is applied.
- `M_antiferro_SU3`: staggered combination formed by the mixed-lattice observable code.

5.4 Recovering the physical Tm dipole

The physically observable Tm magnetic dipole in lab Cartesian components is

$$J_{\mathrm{lab}}^{\alpha}(j) = \eta_{\mu(j),\alpha} \sum_{a\in\{2,5,7\}} \mu_{\alpha a}\, \lambda_a^{\mathrm{loc}}(j). \tag{27}$$

The SU(3) arrays are not themselves the Cartesian dipole; they are the Gell-Mann expectation values in the local CEF basis or after the sublattice-frame map. The physical Tm magnetization is recovered by projecting the T-odd channels with μ_{act} and then applying the Pbnm sign pattern D_μ.

6 Drive Hamiltonian

The THz drive couples to both Fe and Tm via magnetic-dipole and (at Tm) electric-dipole vertices. The microscopic drive terms are (`tmfeo3_notes.tex` Eqs. (Hdrivem)–(Hdrivee))

$$H^{\mathrm{m}}_{\mathrm{drive}}(t) = -g_e\mu_B \sum_i \mathbf{h}(t)\cdot\mathbf{S}_i - g_J\mu_B \sum_j \mathbf{h}(t)\cdot\mathbf{J}^{\mathrm{eff}}_j, \tag{28}$$

$$H^{\mathrm{e}}_{\mathrm{drive}}(t) = -\sum_j \mathbf{e}(t)\cdot\mathbf{d}^{\mathrm{eff}}_j. \tag{29}$$

The two THz propagation geometries are (for propagation along $\hat{b}$):

$$\begin{aligned} &H \,\|\, a,\ E \,\|\, c : h_x \neq 0,\ e_z \neq 0 \ \Rightarrow\ \text{drives } \lambda^{5,7} \text{ and } \lambda^{4,6};\\ &H \,\|\, c,\ E \,\|\, a : h_z \neq 0,\ e_x \neq 0 \ \Rightarrow\ \text{drives } \lambda^{2} \text{ and } \lambda^{1}. \end{aligned} \tag{30}$$

6.1 Pulse envelope

The code supports up to two simultaneously active pulses ($k = 1, 2$) per species. Each pulse carries a Gaussian-modulated cosine envelope:

$$f_k(t) = A \exp\left[-\frac{(t - t_{B_k})^2}{4\sigma^2}\right] \cos\big[\omega(t - t_{B_k})\big], \tag{31}$$

where A (`pump_amplitude`), σ (`pump_width`), and ω (`pump_frequency`) are set independently for the Fe (SU(2)) and Tm (SU(3)) sectors. The intensity profile $I(t) \propto e^{-(t-t_{B_k})^2/(2\sigma^2)}$ has standard deviation σ. A tabulated-waveform mode (via `pump_table_file`) replaces the Gaussian with linear interpolation of a normalized $E(t)$ file.

6.2 Fe drive

The pulse direction for Fe atom type μ is a 3-component Cartesian vector $\mathbf{b}^{(k)}_\mu$, which is transformed to the local sublattice frame before integration:

$$\tilde{\mathbf{b}}^{(k)}_\mu = D^T_\mu\, \mathbf{b}^{(k)}_\mu. \tag{32}$$

The drive contribution to the local effective field on Fe site i is

$$\mathbf{h}^{\mathrm{Fe}}_{\mathrm{drive},i}(t) = \sum_{k=1}^{2} f_k(t)\, \tilde{\mathbf{b}}^{(k)}_\mu, \tag{33}$$

giving the Hamiltonian term $H^{\mathrm{m,Fe}}_{\mathrm{drive}}(t) = -\sum_i \mathbf{h}^{\mathrm{Fe}}_{\mathrm{drive},i}(t)\cdot\mathbf{S}_i$, reproducing the $-g_e\mu_B\sum_i \mathbf{h}\cdot\mathbf{S}_i$ term of Eq. (29) with $A = g_e\mu_B\, h_{\mathrm{peak}}$.

6.3 Tm drive

The pulse direction for Tm atom type ν is an 8-component Gell-Mann vector $\mathbf{c}_\nu^{(k)}$, transformed to the local CEF frame as

$$\tilde{\mathbf{c}}_\nu^{(k)} = F_\nu^T \, \mathbf{c}_\nu^{(k)}. \tag{34}$$

The drive contribution to the local SU(3) effective field on Tm site j is

$$\mathbf{h}_{\text{drive},j}^{\text{Tm}}(t) = \sum_{k=1}^{2} f_k(t) \, \tilde{\mathbf{c}}_\nu^{(k)}, \tag{35}$$

giving $H_{\text{drive}}^{\text{Tm}}(t) = -\sum_j \mathbf{h}_{\text{drive},j}^{\text{Tm}}(t) \cdot \boldsymbol{\lambda}_j$.

Channel assignments for physical fields. Any of the eight Gell-Mann channels may be driven by setting the corresponding component of `pump_directions_su3`. For a physical electromagnetic wave with field components $h_\alpha(t)$ and $e_\alpha(t)$:

- *Magnetic* $h_z \Rightarrow c_2 = g_{\text{ratio}} \mu_{2z}$ $\quad$ [λ^2, A_1^-, drives ω_{12} coherence]
- *Magnetic* $h_x \Rightarrow c_5 = g_{\text{ratio}} \mu_{5x}$, $c_7 = g_{\text{ratio}} \mu_{7x}$ $\quad$ [$\lambda^{5,7}$, A_2^-, drives ω_{13}/ω_{23}]
- *Electric* $e_z \Rightarrow c_4 \propto p_{z4}$, $c_6 \propto p_{z6}$ $\quad$ [$\lambda^{4,6}$, A_2^+, drives ω_{13}/ω_{23}]
- *Electric* $e_x \Rightarrow c_1 \propto p_x$ $\quad$ [λ^1, A_1^+, drives ω_{12} coherence]

The `auto_su3_pump` shortcut. When `auto_su3_pump` is enabled, the runner projects a 3-component physical direction $\hat{\mathbf{e}}$ onto the active T-odd channels via $c_a = \sum_\alpha \mu_{\alpha a} \, e_\alpha$, $a \in \{2, 5, 7\}$. Internally this writes to the 0-based array positions $\{1, 4, 6\}$, which correspond to $\lambda_2, \lambda_5, \lambda_7$ in the 1-based Gell-Mann labeling used throughout this document.